



Course Guidebook for Maternal, Infant & Child Nutrition /HuNu2104

1. GENERAL INFORMATION OF THE COURSE

Course Name:	Maternal, Infant & Child Nutrition /HuNu2104
Course Code:	HuNu2104
Course Credits Points:	3hrs
Contact Hours:	3hrs lecture per week
Course Module:	Nutrition Application II /HuNu-M3121
Course Category:	Major
Status of Course:	Compulsory
Pre-Requisites:	Introduction to Human Nutrition, Nutrition Throughout the Life Cycle Human Nutrition program
Students' Department:	Human Nutrition
Degree Program:	Regular
Students' Admission:	Regular
Year of Study: Semester:	3 rd , First Semester
Academic Year:	2024/2025
Course Duration:	From October 2024 to December 2024
Course Chair:	Adonia Damtew
Course Instructors:	Adonia Damtew, MSc in Human Nutrition Email: adoniadamtew@gmail.com Office: Block 8 Room no.2
Consultation:	Monday morning 10:00AM -11:00AM and Friday afternoon 2:00PM-3:30PM

2. COURSE DESCRIPTION

The course is designed to acquaint students with knowledge of integrated and holistic approach to promote better nutrition during pregnancy, lactation, infancy, and childhood focusing on the nutritional requirements, feeding practices and nutritional interventions related to these groups and also students will have an overview of national and global policies, strategies, programs and services on maternal, infant and child nutrition

3. OBJECTIVES OF THE COURSE

By the end of the course students will be able to

After completion of this course, students will be able:

- To discuss physiological adjustments and nutritional requirements during pregnancy, lactation, infancy and childhood.
- Counsel mothers about the recommended dietary practices during pre-pregnancy, pregnancy, lactation, infancy and childhood
- Promote global and national policies, strategies, programs and services on maternal, infant and child nutrition

4. Application of learning theories with individuals and groups in community settings COURSE Learning OUTCOMES (CLO)

At the end of the course, students will be able to:

No	Course Learning Outcome (CLO)	Performance Indicators (PIs)	Program Student Outcome (PSO)
1	To discuss physiological adjustments and nutritional requirements during pregnancy, lactation, infancy and childhood.	1.1	1
2	Counsel mothers about the recommended dietary practices during pre-pregnancy, pregnancy, lactation, infancy and childhood	2.1,2.2	1
3	Promote global and national policies, strategies, programs and services on maternal, infant and child nutrition	3.1,3.2,3.3	2
4	To discuss physiological adjustments and nutritional requirements during pregnancy, lactation, infancy and childhood.	4.1,4.2,4.3	2
5	Counsel mothers about the recommended dietary practices during pre-pregnancy, pregnancy, lactation, infancy and childhood	5.1,5.2,5.3	3

1. THE COURSE TIME SCHEDULE AND TEACHING METHODOLOGY

The following topics will be covered in this course within this semester.

Table 1. Time Schedule and Teaching Methodology of the course

Time Schedule	Course Contents	Teaching Methodology and References	Attaining student outcome
Week 1,2	1. 1. Introduction 1.1 Why study maternal, infant and child nutrition 1.2 The life cycle approach to nutrition 1.3 The issue of maternal and child nutrition in sub-Saharan Africa	- Two examples should be done. - Lecture, reading, class work, discussions	Outcome 1
Week 2-5	2. Nutrition during Pregnancy and Lactation 2.1 Preconception nutrition 2.2 Nutrition during pregnancy 2.2.1 Physiological adjustments during pregnancy 2.2.2 Energy and nutrient needs during pregnancy 2.2.3 Fetal growth and development 2.2.4 Weight gain during pregnancy 2.2.5 Food beliefs, cravings, avoidances and aversions 2.2.6 Nutrition and adolescent pregnancy 2.3 Nutrition during lactation 2.3.1 Physiological adjustments during lactation 2.3.2 Energy and nutrient needs during lactation 2.3.3 Composition of human milk 2.3.4 Breast milk and formula milk 2.3.5 Indicators for use of breast milk substitutes	- Lecture, Reading, discussions - At least two examples should be done.	Outcome 2

	2.4 Factors that influence maternal and infant outcomes 2.5 Effect of maternal malnutrition on fetus 2.6 Assessment and interpretation of nutritional status of mothers		
Week 6-9	3. Nutrition during Infancy 3.1 Key developmental milestones 3.2 Infant and young child feeding (IYCF) 3.2.1 Breastfeeding 3.2.1.1 Advantages of breastfeeding 3.2.1.2 Breast milk supply and demand 3.2.1.3 Proper Positioning and Attachment 3.2.1.4 Initiation of Breastfeeding 3.2.1.5 Exclusive Breastfeeding 3.2.1.6 Optimal duration of Breast-feeding 3.2.1.7 Maintenance of Lactation during Separation and Special Breast-feeding Circumstances 3.2.1.8 Common Breastfeeding conditions & difficulties 3.2.1.9 Ten steps to successful breastfeeding 3.2.2 Baby friendly hospital initiative 3.2.3 Complementary feeding 3.2.3.1 Timing of complementary feeding 3.2.3.2 Quality and quantity of complementary feeding 3.2.3.3 Optimal complementary feeding practices 3.2.4 Infant and young child feeding during illness 3.2.5 Infant and young child feeding in the context of HIV 3.2.6 Infant and young child feeding in emergency 3.2.7 Growth monitoring and evaluation of infants	▪ Lecture, Reading, discussions ▪ At least two examples should be done.	Outcome 3
Week 10	4. Nutrition of pre-school children 4.1 Growth 4.2 Nutritional requirements 4.3 Parents role 4.4 Food refusal 4.5 Optimal feeding practices	▪ Lecture, Reading, discussions ▪ At least one example should be done	Outcome 4
Week 11-13	5. Nutrition of school aged children 5.1 Growth 5.2 Nutritional requirements 5.3 Adverse reaction to foods 5.4 Childhood obesity 5.5 Optimal feeding practices	▪ At least two examples to be done ▪ Lecture, Reading, discussions	Outcome 5
Week 14-16	6. Harmful traditional practices 7. Essential Nutrition Actions (ENA)	▪ Lecture, Reading, discussions ▪ At least two examples to be done	Outcome 6

	8. Global and national policies, strategies, programs and services on maternal, infant and child nutrition		
Week 17	Final Exam		

2. ASSESSMENT AND EVALUATION

The course utilizes a variety of continuous assessment and evaluation methods to ensure the successful achievement of course learning outcomes. These methods include assignments, projects, quizzes, midterm exams, final exams, site visit reports, homework, classwork, presentations, and more. It is imperative that all assignments and tasks are completed within the specified deadlines.

Home works, Assignments and project: Assignments may involve collaborative work for brainstorming and developing general approaches. However, unless specified otherwise, each individual must complete their own submissions. Assignments and projects may entail both individual and group components, with equal participation expected from all group members. While collaboration is encouraged, submitting another individual's work as one's own constitutes academic dishonesty and will incur in penalties. All group members must contribute equally. Timely submission is mandatory, as late submissions will not be graded. Oral exams will accompany assignments and projects, requiring thorough preparation from all members. Additionally, the instructor reserves the right to assign additional tasks or assignments as necessary.

Quizzes: Quizzes serve to assess students' current understanding of the coursework. They are conducted in class following lecture or tutorial sessions.

Mid-term Exam and Final Exam: The exams will be closed book. During exam necessary materials will be provided by examiner.

Table 2. Summary of Assessments

No	Assessment type	Mark allotted	Content Coverage	Attaining course learning outcome (CLO)	Time Schedule
1.	Assignment 1	5	Chapter 2	1	Week 1
2.	Assignment 2	5	Chapter 3	2	
3.	Quiz Exam 1	10	Chapter 1	3	Week 3
4.	Mini-Project	5	Chapter 3	6	
5.	Mid exam	25%	Chapter 2&3	1,2,3	
6.	Final Exam	50%	Chapter 4,5,6,7&8	4,5,6	Week 12
	Total	100%			

3. LABORATORY/WORKSHOP/ SESSION

Table 3. Summary of laboratory /Work shop/ session content and required material

No	Schedule	Laboratory Topic	Short Description of laboratory	Attaining course outcome
2	Week 3-9	Complimentary feeding practice demonstration, good attachment and positioning,	Interactive lecture, role play, group discussions, brainstorming, case study, drill, role play, video show, poster show, practical teaching: counseling, support mothers in breast feeding and/ Growth Monitoring Chart, Breast Model, Dolls, Posters & Videos	

4. COURSE POLICY

Throughout the duration of this course, all students are expected to follow the student code of behavior as outlined in the “Bahir Dar Institute of Technology Senate Mandate Legislation” or “BiT Senate Mandate Legislation, 2021”. Academic misconduct, such as cheating, fabrication, and plagiarism, will not be permitted and will be quickly reported to the proper authorities for action.

Students who miss more than 15% of the semester's classes will not be eligible to take the final exam. Any student who surpasses the 15% absence level will receive a grade of 'NG' (No Grade) and must present valid reasons for their absence, according to Senate law, Art.37.2 and Art 37.3 (a) and (b) for NG management.

All students are required to have complete records of course assessments and evaluations. If a student misses a quiz, midterm exam, or any other continuous assessment without a valid reason, they will receive a score of zero. However, if a student fails to attend the final exam or complete a term paper/project without a valid reason, the instructor will record an "NG" according to senate legislation, Art. 40.8.

5. GRADING SYSTEM

The grading scale of the course is fixed scale according to senate legislation of Art. 177.

6. TEXTBOOKS

1. Judith E. Brows. (2011). Nutrition throughout the life cycle.4th ed
2. Jones & Bartlett. (2011). Essentials of Life Cycle Nutrition.
3. FMOH (Federal Ministry of Health). 2004. National strategy for infant and young child feeding. Family health department of Ethiopia.
4. PAHO/WHO (Pan American Health Organization, World Health Organization). 2003. Guiding principles for complementary feeding of the breastfed child. Division of Health Promotion and Protection/Food and Nutrition Program, Washington DC, USA
5. M. Nithya Deve. (2012). Handbook of Pediatric Nutrition.
6. Madhu Sharma. (2009). Basic Pediatric Nutrition.

The following bodies confirmed the course guidebook accordingly.

Position	Name	Signature	Date
Course Instructors	Adonia Damtew		
Course Module Chair	Alelegn Amare		
Chair Holder	Mengistu Gizaw		
V/Dean	Behailu B.		